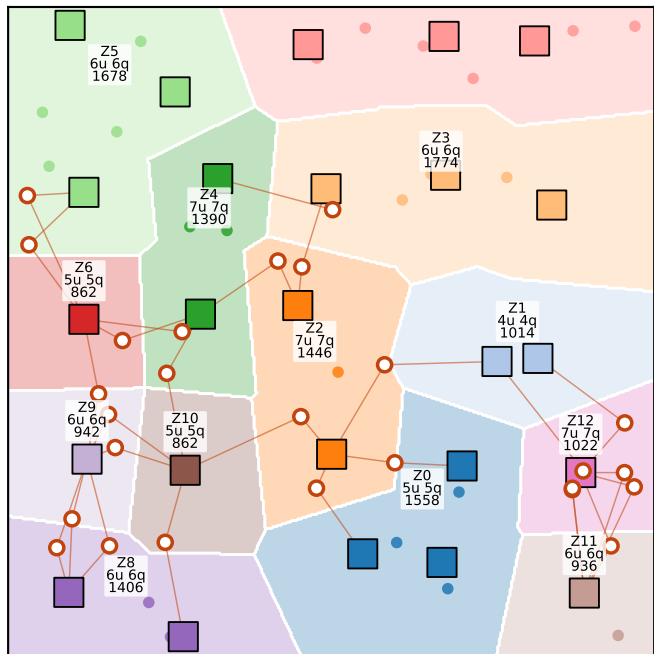


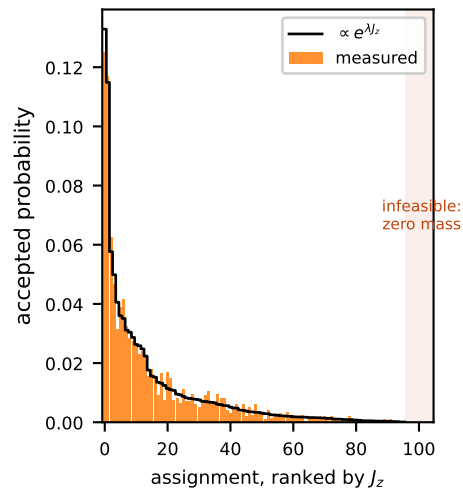
A region partitioned to the hardware budget, solved by zone-local sampling and classical boundary coordination

(a) partition under a 2000 two-qubit budget
25 APs, 50 UEs $\rightarrow$ 13 zones, $|\mathcal{B}|=26$ boundary UEs (52%)

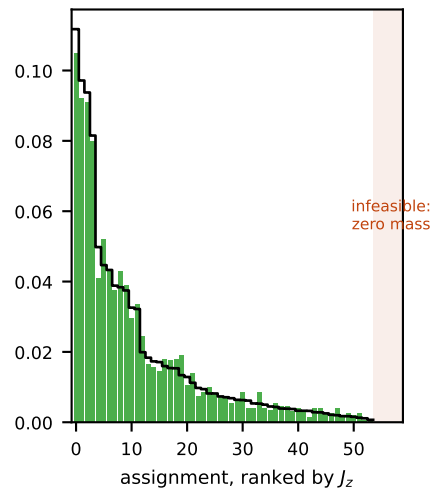


shading: zone territory (nearest AP)
 AP (colour = zone) interior UE
 boundary UE UE-to-candidate-AP link

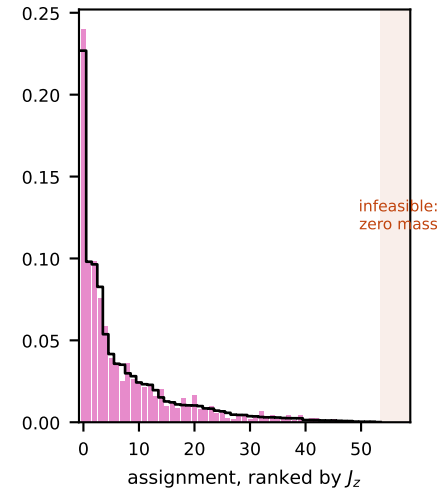
zone Z2: $N_z=7$, $|\mathcal{F}_z|=96$
 $\mu_z=0.026$, $k=4$ rounds



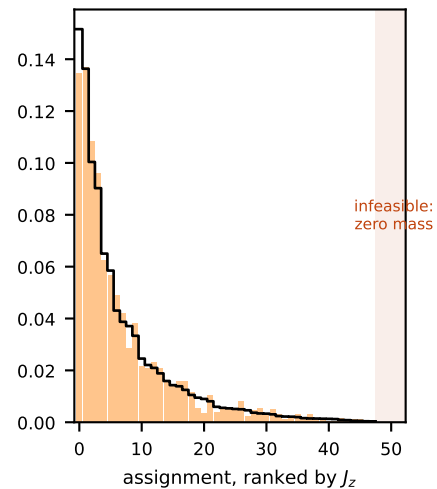
zone Z4: $N_z=7$, $|\mathcal{F}_z|=54$
 $\mu_z=0.015$, $k=6$ rounds



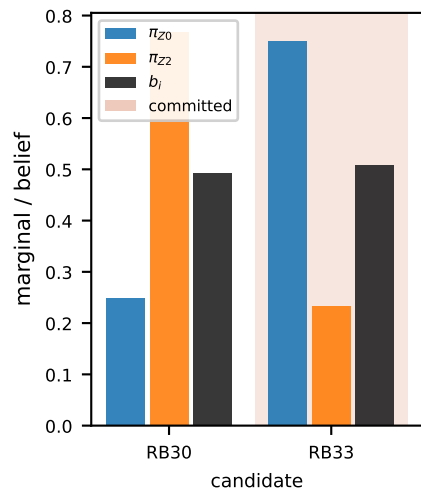
zone Z12: $N_z=7$, $|\mathcal{F}_z|=54$
 $\mu_z=0.001$, $k=26$ rounds



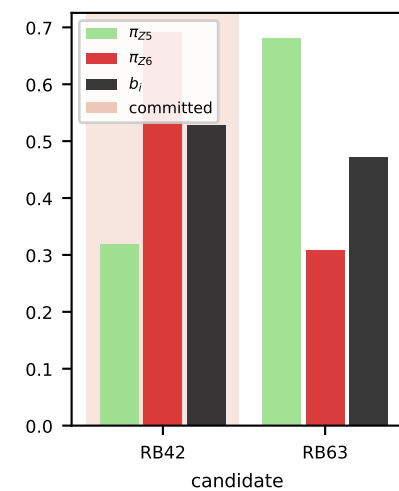
zone Z3: $N_z=6$, $|\mathcal{F}_z|=48$
 $\mu_z=0.103$, $k=2$ rounds



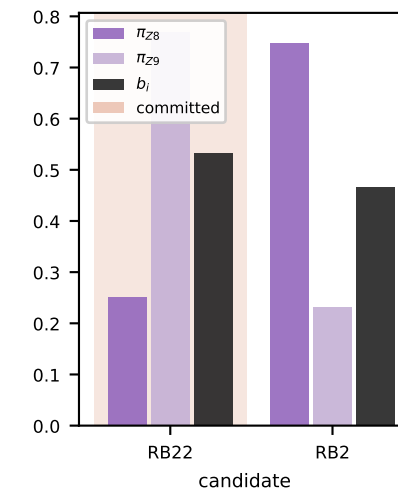
boundary UE 27: Z0 + Z2
confidence 0.51



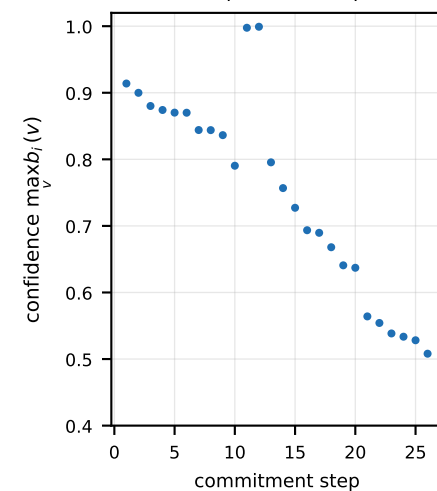
boundary UE 40: Z5 + Z6
confidence 0.53



boundary UE 14: Z8 + Z9
confidence 0.53



decimation order
1 exception re-samples



$g=5$, seed=2 (median of 8 seeds by utility ratio); $\beta=1.5$, $K_z=2000$. Utility 300.5 of the centralized strict optimum 301.0 = 99.8%; the assignment is strictly feasible (True). Zone labels in (a): UEs, state qubits, two-qubit gates. Zone laws are exact: the accepted branch of the circuit matches $\exp(\lambda J_z)$ on $\mathcal{F}_z$ to statevector precision (max TVD $2.7e-15$ over the zones checked in `haiq_certify.py`), so the sampler used here is the same law the circuit produces.